



**THE UNIVERSITY OF AZAD JAMMU & KASHMIR,
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CLI Testing – PC 3

○ USING Command.. ipconfig

```
FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...: 
    Link-local IPv6 Address . . . . .: FE80::202:4AFF:FE5E:109C
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 192.168.1.1
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                0.0.0.0

Bluetooth Connection:

    Connection-specific DNS Suffix...: 
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                0.0.0.0
```

The output successfully verified that the static IP address and subnet mask entered in the graphical interface were properly applied to the PC's network adapter.

○ USING Command.. ping

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

The ping command was used to test network connectivity between PC3 and PC4. Receiving four consecutive "Reply from..." messages confirms that data packets are successfully being exchanged between the two nodes across the switch without any packet loss.

```
C:\>tracert 192.168.1.2

Tracing route to 192.168.1.2 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    192.168.1.2

Trace complete.
```

○ USING

Command.. tracert

The tracert command was run to display the specific path the packets took to reach the destination IP address. The output displays only a single hop because both PCs are directly connected to the same local switch, meaning the data does not need to traverse multiple routers to reach its target.

○ USING command.. arp -a, netstat

```
C:\>arp -a

Internet Address      Physical Address      Type
192.168.1.2           000d.bd79.6105       dynamic
```

```
C:\>netstat

Active Connections

Proto Local Address          Foreign Address         State
C:\>
```

The
arp -a

command successfully displayed the device's Address Resolution Protocol (ARP) table, which maps the known logical IP addresses to their corresponding physical MAC addresses.

Additionally, the netstat command was executed to display active network connections and statistics for the system.

CLI Testing – PC 4

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...: 
    Link-local IPv6 Address . . . . .: FE80::20D:BDFE:FE79:6105
    IPv6 Address . . . . .: ::
    IPv4 Address. . . . .: 192.168.1.2
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                   0.0.0.0

Bluetooth Connection:

    Connection-specific DNS Suffix...: 
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address. . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                   0.0.0.0
```

○ Using command.. ipconfig

The output successfully verified that the static IP address and subnet mask entered in the graphical interface were properly applied to the PC's network adapter.

○ USING command.. ping

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

The ping command was used to test network connectivity between PC3 and PC4. Receiving four consecutive "Reply from..." messages confirms that data packets are successfully being exchanged between the two nodes across the switch without any packet loss.

```
C:\>tracert 192.168.1.1

Tracing route to 192.168.1.1 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    192.168.1.1

Trace complete.
```

○
Using

command.. tracert

The tracert command was run to display the specific path the packets took to reach the destination IP address. The output displays only a single hop because both PCs are directly connected to the same local switch, meaning the data does not need to traverse multiple routers to reach its target.

○ USING command.. arp -a, netstat

```
C:\>arp -a

Internet Address      Physical Address      Type
192.168.1.1           0002.4a5e.109c        dynamic
```

```
C:\>netstat

Active Connections

Proto Local Address      Foreign Address    State
C:\>|
```

The arp -a command successfully displayed the device's Address Resolution Protocol (ARP) table, which maps the known logical IP addresses to their corresponding physical MAC addresses. Additionally, the netstat command was executed to display active network connections and statistics for the system.
